

TWO HILLS AGDM, AB, Canada

WMO: 712760

Lat: 53.6253N

Lon: 111.6783W

Elev: 2224

StdP: 13.55

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.3	(c) -17.6	(d) -29.8	(e) 1.1	(f) -23.0	(g) -23.4	(h) 1.6	(i) -17.3	(j) 30.2	(k) 24.5	(l) 27.1	(m) 20.6	(n) 8.4	(o) 160	(p) 0.728

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.3	(c) 82.2	(d) 62.8	(e) 78.9	(f) 61.7	(g) 75.8	(h) 60.6	(i) 66.6	(j) 77.8	(k) 64.5	(l) 75.0	(m) 62.5	(n) 72.3	(o) 10.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 62.4	(b) 91.7	(c) 72.2	(d) 60.3	(e) 84.8	(f) 69.5	(g) 58.4	(h) 79.1	(i) 67.5	(j) 32.5	(k) 78.4	(l) 30.8	(m) 75.0	(n) 29.2	(o) 72.3	(p) 75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	22.6	20.1	-29.4	87.8	5.4	3.1	-33.3	90.0	-36.5	91.8	-39.6	93.5	-43.5	95.7
			-29.6	71.3	5.4	2.1	-33.5	72.8	-36.7	74.0	-39.7	75.1	-43.7	76.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	10.5	11.3	23.0	38.6	51.2	58.4	62.5	60.2	51.7	39.0	22.7	11.6
	DBStd	22.00	15.42	13.71	12.48	9.65	8.33	5.72	5.24	6.00	8.34	9.13	12.79	13.67
	HDD50	6039	1225	1083	838	358	85	7	0	3	76	354	820	1190
	HDD65	10346	1690	1503	1303	793	431	210	112	170	404	805	1270	1655
	CDD50	1247	0	0	0	15	124	258	388	319	128	15	0	0
	CDD65	77	0	0	0	0	4	11	35	21	6	0	0	0
	CDH74	1064	0	0	0	2	91	147	409	280	132	3	0	0
	CDH80	204	0	0	0	0	15	19	93	45	32	0	0	0
	WSAvg	10.0	10.3	9.7	10.4	11.3	11.0	10.0	8.8	8.7	9.7	10.4	10.0	9.7
	PrecAvg	15.5	0.7	0.4	0.6	1.1	1.7	2.6	2.9	2.2	1.3	0.7	0.7	0.7
(15) Precipitation	PrecMax	19.6	2.3	0.9	1.5	4.1	4.0	5.4	5.9	4.8	3.0	1.9	3.1	1.2
	PrecMin	11.4	0.1	0.0	0.2	0.1	0.2	0.5	0.9	0.4	0.0	0.0	0.1	0.1
	PrecStd	2.6	0.5	0.2	0.3	0.8	0.9	1.1	1.1	0.9	0.8	0.4	0.7	0.3
	PrecStd	2.6	0.5	0.2	0.3	0.8	0.9	1.1	1.1	0.9	0.8	0.4	0.7	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	41.1	41.4	52.9	71.6	82.2	82.7	87.3	84.7	84.5	72.0	53.7	42.1	
	0.4% MCWB	35.8	36.1	43.1	50.8	57.9	61.9	66.3	63.5	60.6	53.5	43.4	36.8	
	2% DB	38.2	37.2	44.8	65.2	75.9	78.4	82.3	80.3	77.7	62.5	46.1	37.8	
	2% MCWB	34.3	33.3	38.2	47.8	54.7	60.1	65.7	63.0	58.0	49.4	39.4	33.7	
	5% DB	35.5	33.9	40.8	59.0	71.4	74.3	78.5	77.1	72.2	56.8	41.3	34.3	
	5% MCWB	32.4	30.8	35.8	44.7	52.9	58.6	64.4	62.4	56.1	46.0	35.8	30.7	
	10% DB	31.8	30.7	37.7	54.0	67.2	70.7	74.9	73.5	66.2	52.4	37.5	30.3	
	10% MCWB	29.2	28.3	33.7	42.5	50.6	56.5	62.9	60.9	53.7	43.7	33.7	27.7	
	0.4% WB	36.6	36.6	43.5	52.2	59.7	65.5	70.9	68.1	62.0	54.3	44.5	37.2	
	0.4% MCDB	40.4	40.8	51.4	67.6	73.6	77.6	80.8	77.8	78.5	69.8	52.3	41.5	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.7	33.5	38.6	48.5	56.5	62.4	67.7	65.6	59.1	50.6	40.0	34.1	
	2% MCDB	37.9	36.3	44.0	63.3	70.9	73.4	78.6	76.4	74.3	61.0	45.7	37.8	
	5% WB	32.5	31.1	36.0	45.5	54.3	60.5	65.6	63.4	56.8	47.1	36.6	30.9	
	5% MCDB	35.6	33.8	40.2	57.7	68.6	70.5	76.5	73.9	70.1	55.0	40.4	33.9	
	10% WB	29.1	28.1	34.0	43.0	52.2	58.5	63.7	61.5	54.5	44.5	33.9	27.8	
	10% MCDB	31.7	30.7	37.4	53.0	65.1	68.0	73.3	71.6	65.0	51.4	37.1	30.4	
(35) Mean Daily Temperature Range	MDBR	15.1	15.5	15.2	18.4	21.6	19.3	20.3	21.0	20.9	17.3	13.7	13.7	
	5% DB	16.0	18.5	17.4	26.8	27.3	24.9	25.7	26.8	30.3	25.1	17.5	17.2	
	5% WB	13.3	15.5	12.2	13.9	12.0	11.2	12.6	12.7	13.5	14.1	12.0	14.2	
	MCDBR	15.9	17.9	16.5	24.9	24.1	21.4	23.4	23.8	27.3	23.6	16.5	17.1	
	MCWBR	13.3	15.3	11.8	13.7	11.3	11.0	12.5	12.4	12.8	13.8	12.0	14.5	
	MCWBR	13.3	15.3	11.8	13.7	11.3	11.0	12.5	12.4	12.8	13.8	12.0	14.5	
(40) Clear-Sky Solar Irradiance	taub	0.248	0.271	0.301	0.328	0.346	0.361	0.361	0.371	0.309	0.275	0.249	0.225	
	taud	2.309	2.299	2.323	2.370	2.371	2.354	2.384	2.326	2.489	2.553	2.444	2.342	
	Ebn at Noon	236	266	281	287	286	281	279	269	277	267	239	227	
	Edh at Noon	18	25	31	34	36	37	35	35	26	20	16	14	
(44) All-Sky Solar Radiation	RadAvg	305	639	1120	1485	1839	1886	1941	1586	1112	638	305	225	
	RadStd	33	47	88	130	133	134	104	116	120	89	34	22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page